Pool-ball localisation and recognition

Manon Yiqin Galian Ng (23375206) — CSU44053

1 Processing and design

The system finds and identifies balls, then shows their positions from above. AllBalls supplies detection settings and reference colours; Tables1–6 assess the result. Separate detection and recognition stages help locate errors.

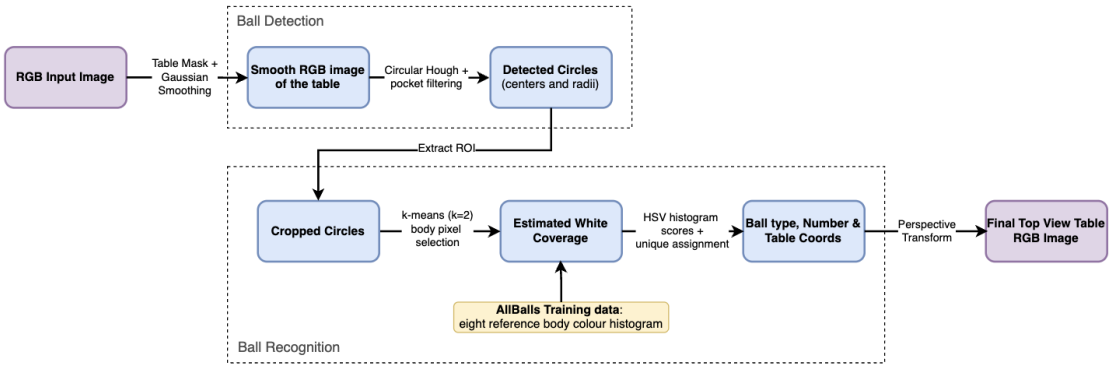


Figure 1: Final pipeline. AllBalls supplies the reference colour histograms.

1.1 Table masking and circular Hough detection

The supplied corners define a mask excluding objects outside the table. Greyscale conversion and smoothing precede the circular Hough transform [1]. A 5×5 Gaussian with $\sigma = 1.5$ suppresses small intensity variations while preserving outlines. These fixed smoothing settings were not optimised. Circles are detected before perspective correction to preserve their outlines.

Manual tuning traded missed balls against false circles. I therefore compared 480 Hough configurations on AllBalls, preferring more matches, fewer false detections, then lower centre error. The chosen accumulator resolution ratio is 1.5, centre spacing 18 pixels, edge threshold 75 and vote threshold 35. Fixed radii of 15–30 pixels cover the observed sizes.

Pockets also have circular edges. Removing centres within 25 rectified table units of any pocket eliminates an extra AllBalls circle, but risks rejecting genuine balls near pockets.

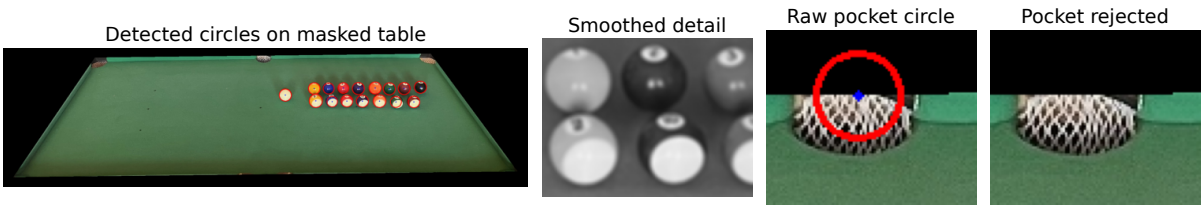


Figure 2: AllBalls: detections on the masked table, smoothing detail and pocket filtering. The 17 raw circles become 16.

1.2 Ball crops and type estimation

Each circle gives a square crop, or region of interest (ROI), for inspecting one ball. A circular mask excludes the square's corners, preventing surrounding cloth from entering recognition. K-means groups the retained pixels into two BGR colour clusters, intended to separate light areas from coloured or dark material. Pixel positions are not clustering features; the CCA experiment later tests whether spatial layout provides useful extra information.

Each cluster's mean is converted to HSV. It is light when $V \geq 170$ and $S \leq 165$; a non-light cluster with $V \leq 90$ is dark. Cluster sizes estimate the light fraction w. White requires $w \geq 0.85$; Solid requires at least 70% non-light pixels; remaining mixtures are Striped. These are the original manually chosen thresholds retained by the K=2 pipeline, rather than the winner of the older recognition search. Clustering uses three initialisations, a fixed seed, and stops after 50 iterations or centre change below 0.2.

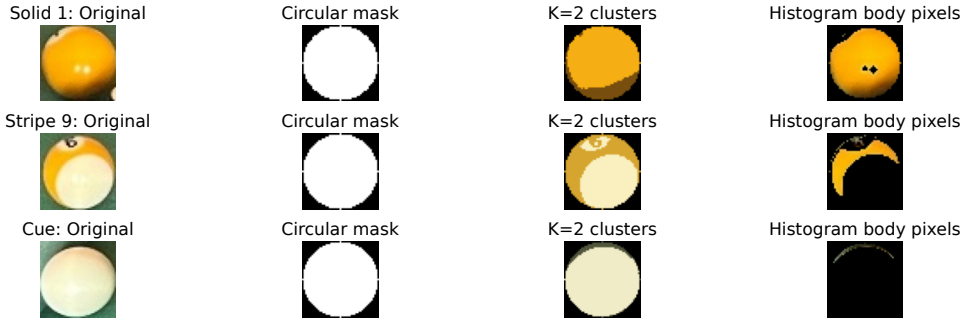


Figure 3: Solid, striped and cue crops: circular mask, K-means segmentation and pixels used for the colour histogram.

1.3 Reference colour histograms

White coverage distinguishes types but cannot identify body colour. I therefore compare colour distributions with eight references: yellow, blue, red, purple, orange, green, maroon and black. AllBalls detections are matched to its annotations to label the reference crops. Solid/striped partners share a colour; the cue is excluded because it has no numbered body colour.

A 3×3 elliptical erosion removes a thin rim that may include cloth. Bright, weakly saturated pixels are excluded using the light thresholds above, reducing white stripes, number patches and reflections. Fewer than 12 remaining pixels gives no usable body-colour evidence. This mask uses individual pixel colours, rather than K-means cluster means.

Each HSV histogram divides hue into 18 bins, saturation into 8 and brightness into 8: $18 \times 8 \times 8 = 1152$ combinations. Nearly grey pixels have unstable hue: if their largest and smallest colour channels differ by less than 15, hue is set to zero. Smoothing replaces bin values with a weighted average of that bin and its neighbours: $1, 2, 1$ divided by four, applied along each axis. Hue wraps around because its endpoints are neighbouring colours. This reduces sensitivity to small colour changes near bin boundaries. Histograms sum to one, allowing differently sized crops to be compared; reference examples are averaged equally.

1.4 From colour similarity to identity scores

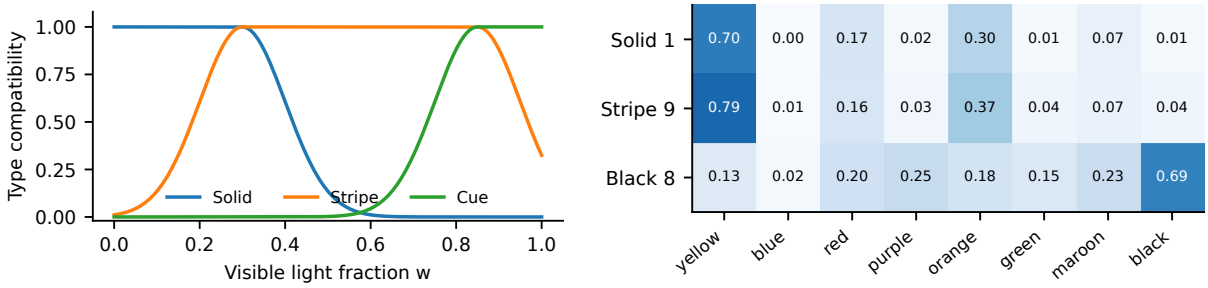
A crop's histogram is compared with each reference using one minus the Bhattacharyya distance. Similar distributions score nearer one; dissimilar ones score nearer zero. The eight scores express colour similarity, not probabilities, and do not need to sum to one. A yellow solid and yellow stripe can therefore both favour yellow strongly while still needing different numbers.

A hard type decision discards alternatives when estimated white coverage is misleading. Instead, final recognition keeps support for all three types. A solid is favoured for $0 \leq w \leq 0.30$, a stripe for $0.30 \leq w \leq 0.85$, and the cue for $0.85 \leq w \leq 1$. Outside these intervals, support decreases gradually rather than suddenly becoming zero.

For type t, let d be the distance from w to its preferred interval: zero inside it, otherwise the gap to the nearest endpoint. I used

$$g_t(w) = \exp\left(-\frac{d^2}{2(0.10)^2}\right).$$

The width 0.10 controls tolerance: a gap of 0.10 gives support about 0.61. This Gaussian-shaped rule is a heuristic, not a learned probability. A numbered identity's score multiplies its colour similarity by its type support. For example, 1 and 9 share yellow similarity but use Solid and Striped support respectively. The cue score is w multiplied by White support.



Calculated type support

Measured colour similarities

Figure 4: Soft type support and example histogram scores. Yellow partners share a colour preference but differ in visible white coverage.

1.5 Unique identity assignment

Choosing every crop's best score independently can give two balls the same number. Since each identity occurs at most once on a table, the Hungarian method [2] chooses the combination with the highest total score while using each identity at most once. Separate unassigned choices allow several Unknowns.

A crop needs usable body-colour evidence or a White classification to claim an identity; scores at most 10^{-9} are ineligible. This weak rejection rule can still let a false detection take a real identity. Uniqueness resolves duplicate claims, but cannot guarantee that the selected numbers are correct.

1.6 Handling apparent duplicate cues

Hough can detect only a stripe's white patch, creating an apparent second cue. Initially White crops expand by one radius pixel and are reanalysed to expose surrounding colour. Original detection geometry is kept. With several White candidates, the largest original radius reserves cue identity: a whole cue usually exceeds a patch despite depth differences. A nearby patch could exceed a distant cue; expected size at each depth would improve this comparison.

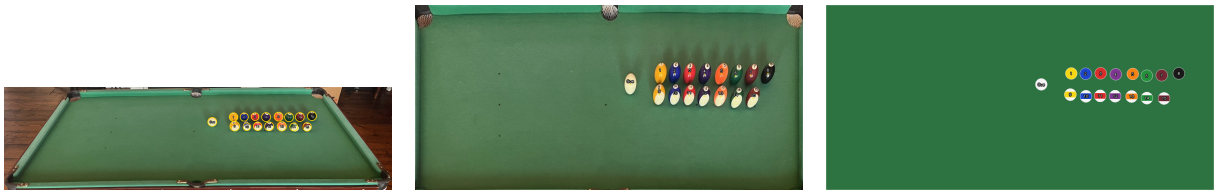


Figure 5: Table3: full cue versus the smaller white patch on ball 12. Centre/right: original and one-pixel-expanded crops, shown at the same pixel scale.

1.7 Table coordinates and final representation

Perspective correction maps the supplied corners to a 1434×683 rectangle.

Recognition uses original-image crops; transformed centres give overhead positions and coordinates for evaluation. The diagram draws predicted identities at those positions, providing a clear view of the result.



Original labels Overhead labels Identity diagram

Figure 6: AllBalls in the three output views; all 16 identities are correct.

1.8 Evaluation protocol

Detections and annotations are paired one-to-one within 15 rectified table units, maximising matches. Matches are true positives (TP), extras false positives (FP), and misses false negatives (FN). Precision is matches/detections, recall is matches/annotations, and F1 their harmonic mean. Mean error $\bar{e} = M^{-1} \sum_{i=1}^{M} \|\hat{\mathbf{x}}_i - \mathbf{x}_i\|_2$ averages Euclidean distance between the M matched detected and annotated centres.

Recognition accuracy divides correct labels by matched balls; end-to-end uses all annotations. Unknown is incorrect. Extra detections reduce precision, but not directly end-to-end accuracy. Colour excludes cues; pooled results exclude AllBalls. Development used Tables1–6, so these are validation images.

2 Results and analysis

2.1 Final performance on Tables1–6

The system finds 86 of 90 annotated balls, with two false detections. Detection precision is **97.73%**, recall **95.56%** and F1 **96.63%**. Final identity accuracy is **93.02%** among matched balls, or **88.89%** end-to-end.

Table 1: Per-image detection and final identity results. Error is in table units.

Image	Balls	TP	FP	FN	Error	Correct IDs
Table1	16	14	1	2	7.68	11/14
Table2	16	16	1	0	3.58	16/16
Table3	16	15	0	1	3.12	15/15
Table4	15	14	0	1	3.89	14/14
Table5	14	14	0	0	5.83	13/14
Table6	13	13	0	0	5.76	11/13
Overall	90	86	2	4	4.91	80/86

The overall centre error, 4.91 table units, averages all 86 matched centre errors, so every matched ball contributes equally.

Table 2: Recognition at each stage.

Stage	Correct / matched	Correct / annotated
Initial type	76/86 (88.37%)	76/90 (84.44%)
Colour, excluding cue	73/80 (91.25%)	73/84 (86.90%)
Final identity	80/86 (93.02%)	80/90 (88.89%)



Figure 7: Final labelled outputs for Tables1–6.

2.2 Analysis of the main failures

Final identity also uses soft scores, cue handling and uniqueness, so it can correct initial appearance errors. These totals do not isolate each addition's benefit. All six cue identities are correct.

Table2 labels every real ball but keeps an extra circle as Unknown. Tables3 and 4 label all matches correctly but each miss ball 10; Tables5 and 6 find every ball but make recognition errors.

Table1 misses 6 and 7, leaves 14 Unknown, exchanges 5 and 9, and gives a false circle number 6. The exchanged matches have centre errors 14.63 and 14.81, near the 15-unit limit. A poorly centred crop can still count as detected while missing stripe pixels or admitting a neighbour, changing recognition evidence.

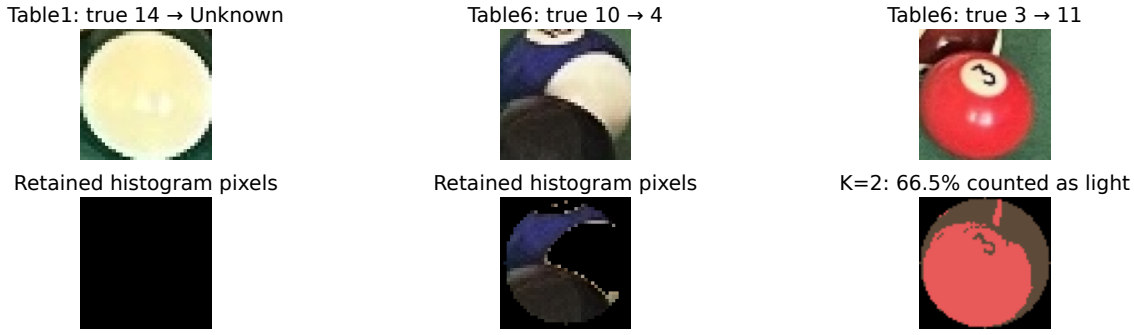


Figure 8: Different recognition failures: too few body pixels, misleading retained colour and a bright red cluster counted as light. Black means excluded pixels.

Too little colour. Table1 ball 14 is detected mainly through its white patch. Even after expansion, too few body pixels remain for a colour histogram, so the final label stays Unknown. Guessing from white coverage alone would not identify its body colour.

Misleading colour evidence. Table6 ball 10 scores 0.52 against black, 0.43 against purple and only 0.24 against blue. The retained pixels include a substantial dark area. Assignment gives it 4 after considering the other balls; its number is therefore affected by both a poor colour description and competition for unique identities. It is not simply choosing the largest colour score.

The light rule accepts bright red. For Table6 solid 3, the main cluster has $S = 157$, $V = 232$, satisfying the light thresholds despite its red colour. It covers 66.5% of the crop. Red remains the strongest histogram match, but the estimated light fraction favours striped red 11. This directly exposes a weakness of the cluster-mean light rule, rather than a failure to recognise red.

The failures span crop geometry, pixel selection and type interpretation. Uniqueness resolves duplicate claims but can also give a false circle an unused number or spread an incorrect claim to a genuine ball.

2.3 Experiments beyond the final pipeline

The following exploratory settings were not exhaustively tuned; their results concern these configurations.

K=2 with CCA. K-means ignores position; connected-component analysis tests whether white regions distinguish stripe bands from small highlights. Eight-connected regions below 2% of ball area are ignored, and a 5×5 erosion defines an inner edge band. A stripe needs a light region covering 20% of the ball, light on 20% of the band and 10% non-light area. These rules aim to exclude tiny highlights, but cues and number patches can also satisfy them. Correct types fall from 76/86 to 73/86: no corrections and three new errors.

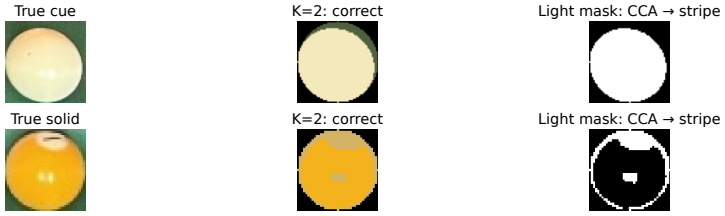


Figure 9: CCA incorrectly changes a cue and solid to Striped.

Guided Hough with optional watershed. Crowded regions extending to 1.4 times detected radii guide local searches. Lower vote threshold 22 and centre spacing 12 pixels admit weaker circles. HSV cloth rejection and 3×3 closing supply foreground. Watershed uses centres as markers, accepting new circles whose foreground support is 0.15–2 times their disk area. These loose area bounds reject unsupported proposals. Duplicates and pocket circles are removed; baseline circles remain.

Table 3: Pooled detection and final recognition with the same recognition pipeline.

Detector	TP	FP	FN	Correct IDs	False numbered
Baseline	86	2	4	80/90	1
Guided Hough	89	8	1	81/90	6
Guided + watershed	88	6	2	80/90	5

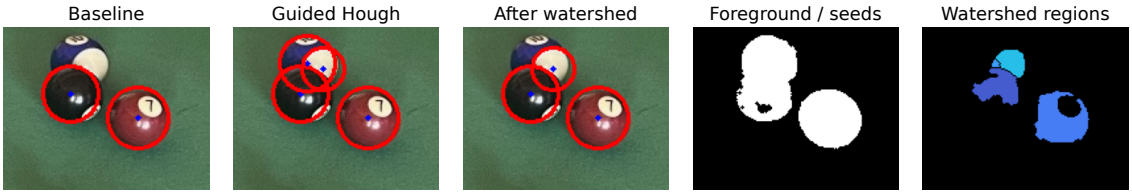


Figure 10: Table3: guided Hough recovers ball 10 and adds a false circle; watershed rejects the recovery. Last panels: foreground/seeds and watershed regions.

Guided Hough recovers three balls but adds six false detections: one net identity gain. Watershed removes that gain. Table3 shows both outcomes and a false circle taking identity 12. The simpler baseline avoids more false numbered outputs; better segmentation or tuning could change this outcome.

2.4 Group performance, robustness and improvements

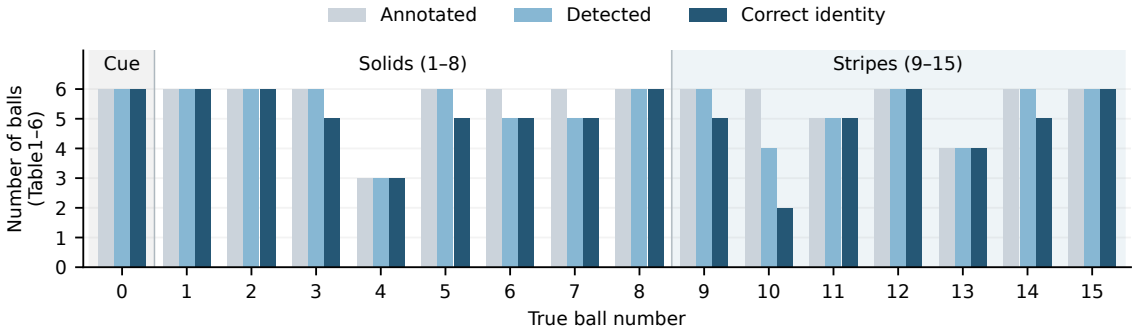


Figure 11: Counts by true identity across Tables1–6: annotated, detected and correctly numbered balls. The vertical axis counts balls, not percentages.

Blue stripe 10 is weakest: six annotated examples give four detections and two correct identities (33.3% end-to-end). Its errors occur at both detection and recognition. Solids give 41/45 correct identities (91.1%), stripes 33/39 (84.6%) and cues 6/6. Stripes depend on how much white and colour the crop captures; orientation, occlusion and displaced circles can therefore alter both type and colour evidence. Some perfect identities have only three or four examples, so their scores are limited evidence of generalisation.

Robustness to scene changes. Circular Hough suits visible ball outlines, histograms tolerate variation within a colour, and uniqueness uses the table's physical constraint. However, fixed corners, radii and pocket zones depend on the camera. A new viewpoint or resolution needs updated geometry and size ranges. Lighting affects histogram scores and the light rule; occlusion removes outline and stripe evidence. The largest-white-radius heuristic can also fail when a nearby patch exceeds a distant cue.

Limits of the evaluation. Six repeatedly inspected images cannot establish performance on new games. One correct identity changes end-to-end accuracy by 1.11 percentage points. The next evaluation should freeze the settings and use new photographs, initially keeping the same camera but varying ball orientation, crowding and lighting. Stricter matching tolerances should also be tested; centre annotations do not measure segmentation accuracy.

Targeted improvements. Priorities are cleaner crowded-ball crops, a light rule that rejects bright coloured clusters, and references with more varied appearances. Expected radius by depth could improve cue selection. A validated Unknown threshold could stop weak false circles taking identities. CCA needs more selective spatial rules; guided watershed needs better foreground and markers. Improvements should be judged on both corrected and newly wrong identities, including false numbered outputs, rather than on segmentation appearance alone.

References and assistance [1] K. Dawson-Howe, CSU44053 Computer Vision course material, Trinity College Dublin.

[2] H. W. Kuhn, "The Hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, 2, 83–97 (1955).

ChatGPT/Codex assisted with programming, diagnostics and report drafting.
Code: `https://github.com/galiannm/pool-ball-recogniser`